# AI-Assisted Kidney Stone Detection & Classification Report

Research Examination ID: 46324E0A | Date: 2026-10-26 20:48:27

## 1. Image & Study Metadata

| Parameter | Value |
|---|---|
| Examination ID | 46324E0A |
| Modality | CT Scan |
| Image Format | JPEG |
| DICOM Calibrated | No (Standard 2D Image) |

## 2. Individual Deep Learning Classification Models

| Model Architecture | Status | Predicted Class | Confidence | Stone Prob. | Non-Stone Prob. | Latency |
|---|---|---|---|---|---|---|
| MOBILENET V2 | Trained & Loaded | **Stone** | 99.5% | 99.5% | 0.5% | 30.01 ms |
| RESNET50 | Trained & Loaded | **Stone** | 99.7% | 99.7% | 0.3% | 55.03 ms |
| EFFICIENTNET B0 | Trained & Loaded | **Stone** | 99.3% | 99.3% | 0.7% | 37.91 ms |

## 3. Multi-Model Ensemble Analysis

| **Ensemble Decision** | **Stone** (99.5% confidence) |
|---|---|
| Stone Probability | 99.5% |
| Non-Stone Probability | 0.5% |
| Contributing Classifiers | mobilenet_v2, resnet50, efficientnet_b0 |
| Normalized Model Weights | {'mobilenet_v2': 0.3333, 'resnet50': 0.3333, 'efficientnet_b0': 0.3333} |
| Model Disagreement | Full Consensus |

## 4. YOLOv8 Object Detection (Localization)

Detection model not trained or unavailable. Stone localization bounding-box annotations required.

## 5. Physical Measurements & Calibrations

**Status:** Uncalibrated 2D ROI: 30 x 25 pixels

• Physical millimeter scale unavailable: Standard PNG/JPEG image does not retain DICOM PixelSpacing metadata.

• Reporting pixel dimensions only. Do NOT make clinical decisions based on uncalibrated pixel sizes.

**MANDATORY MEDICAL & RESEARCH DISCLAIMER:**
This report was generated by an academic research artificial intelligence system. The classification outputs, probability scores, and ensemble results are algorithmic estimates and do NOT constitute a medical diagnosis, clinical radiologist verification, or treatment recommendation. Model confidence

must not be interpreted as an absolute probability of clinical disease. Clinical diagnosis requires thorough review by a qualified radiologist or physician.